

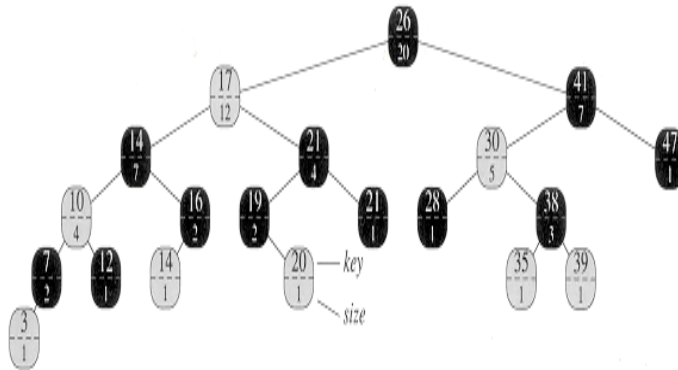
INHA UNIVERSITY IN TASHKENT (FALL 2024)
Computer Algorithms (CIE3090)
ASSIGNMENT #4

Section No. _____ ID _____ Name _____

Submission Deadline: **15 /Nov/ 2024 (23:30 Hr)**

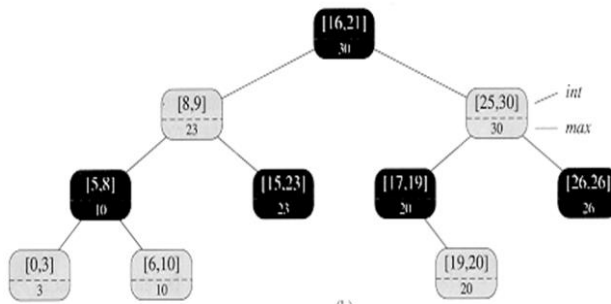
➤ Solve all questions on printout of this page. You are required to solve (**handwritten**) all the questions.

Q1. Find the rank of key 47 in the following order-statistics tree



Iteration	Key	r

Q2. Find the interval [24, 26] in the following Interval tree



Intrvl	maxL	overlap/Move

Q3. Following RB Tree is augmented to obtain Order-Statistic tree.

- Redraw the tree showing the size field in each node.
- Insert the node with key value 30, and restructure the tree

	<u>Draw order statistic tree here with size filed</u>
<u>Insert Key value 30</u>	<u>Final Tree</u>

Below space can be used showing any rotation or adjustment